PSOC™ Edge Protect Bootloader training manual

About this document

Scope and purpose

This document is a training manual for the hands-on exercise with an Edge Protect Bootloader on PSOC™ Edge using the ModusToolbox™.

Intended audience

The intended audiences for this document are design engineers, technicians, and developers of electronic systems.

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1 Introduction

This manual provides specific instructions to create, configure, build, and run the code examples on the PSOC™ Edge E84. In this lab session, you will learn how to use Edge Protect Bootloader on PSOC™ Edge and enable the chain of trust with EPB.

2 Required development tools

- ModusToolbox™ software v3.6 or later
  - Recommended installation via ModusToolbox™ Setup tool
- Eclipse IDE for ModusToolbox™ v2025.4.0 or later
  - Recommended installation via ModusToolbox™ Setup tool
- Edge Protect Security Suite v1.6 or later.
  - Installed by ModusToolbox™ Setup tool as a dependency to ModusToolbox™ v3.6
- ModusToolbox™ Programming Tools v1.5.0 or later
  - Installed by ModusToolbox™ Setup tool as a dependency to ModusToolbox™ v3.6
- Board support package (BSP):
  - KIT_PSE84_EVAL_EPC2: v1.0.0 or later (available with ModusToolbox™ v3.6)
- Terminal emulator
  - Instructions in this document use Tera Term
- PSOC™ Edge E84 Evaluation Kit (KIT_PSE84_EVAL)
  - Ensure device boots from external memory:
- BOOT SW/SW6 in 'ON/HIGH' position
  - Disable alternate serial interface configuration:
    - J20/J21 in Tristate/Not-Connected (NC) position

3 Prerequisites

Install the software and obtain the hardware listed in the **Error! Reference source not found.** section.

This class will not cover basic concepts of ModusToolbox™ and PSOC™ Edge.

- For an introduction to PSOC™, including a getting started guide to ModusToolbox™, go to:

https://www.infineon.com/product-information/psocdeveloper

- For PSOC™ Edge trainings, from beginner tutorials to advanced trainings, go to:

PSOC™ Edge E84 Training Collection

4 Basic Secure app with Edge Protect Bootloader (EPB)

4.1 Objective

Designed for beginners, this lab session provides a step-by-step approach to start using the Edge Protect Bootloader along with a simple application (Basic secure application code example).

4.2 Description

This code example demonstrates the flow as shown in Figure 1.

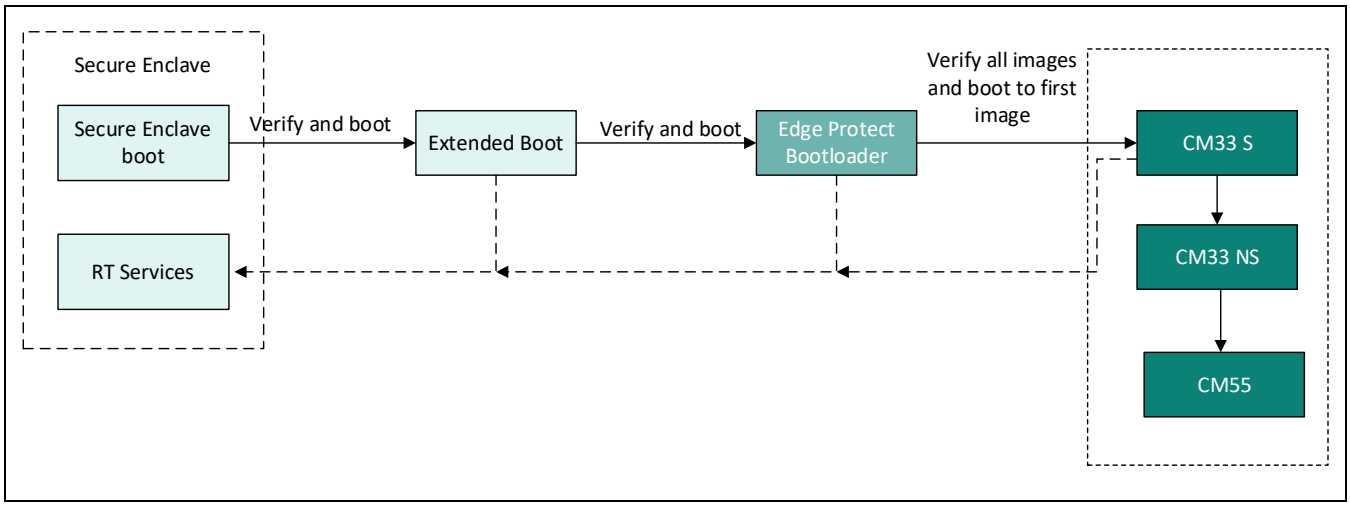


Figure 1 Boot sequence with Edge Protect Bootloader

For more information on the terminology used, refer to the AN237849: Getting Started with PSOC™ Edge Security.

4.3 Project Creation

1. Before creating the first project, we need to create our workspace in ModusToolbox™. Open **ModusToolbox™** from the **Windows Start** menu

Note: If you have not installed ModusToolbox™, see Required development tools . If you have already created a workspace, skip to step 4 directly.

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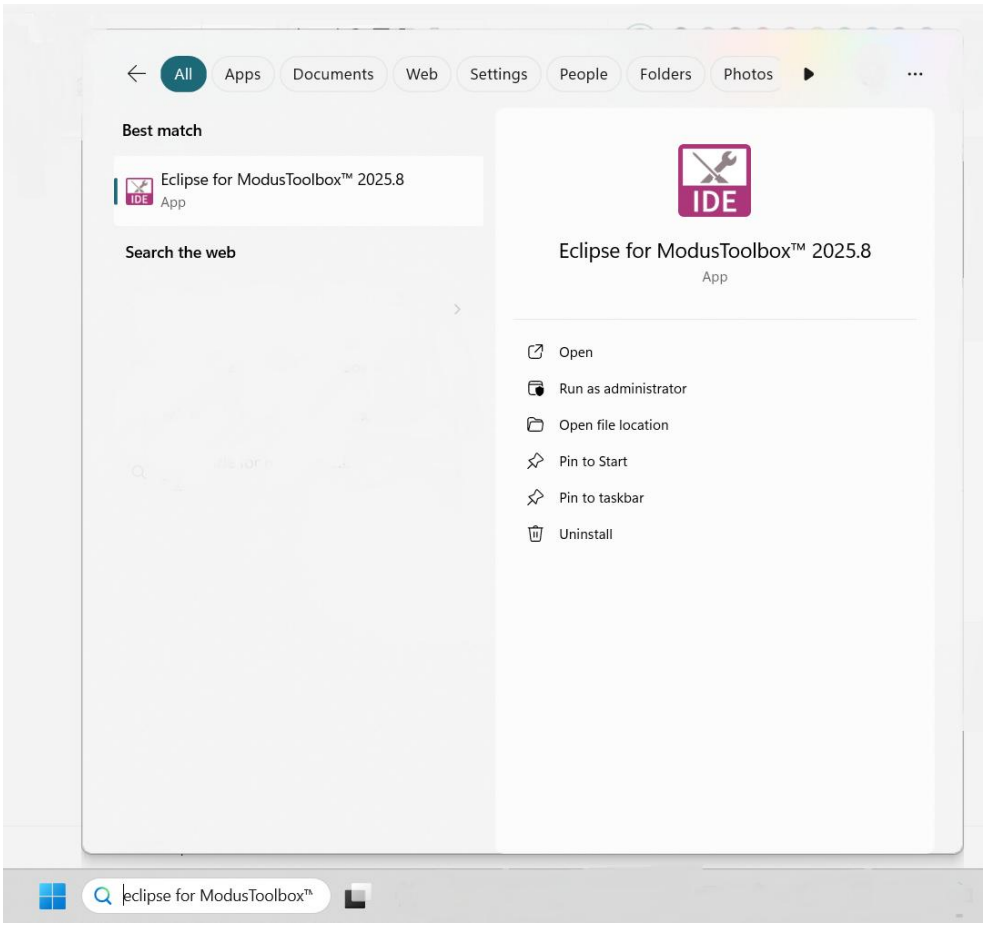


Figure 2 Start ModusToolbox™

2. **Next**, choose a **workspace directory** for your project. After selecting a suitable directory, click the **Launch** button

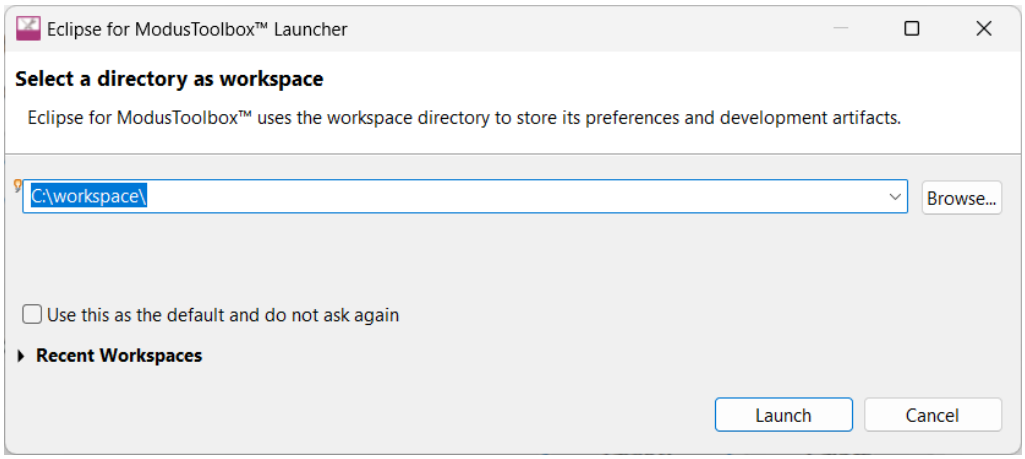


Figure 3 Select workspace directory

3. Launch opens a new ModusToolbox™ workspace as shown in Figure 4

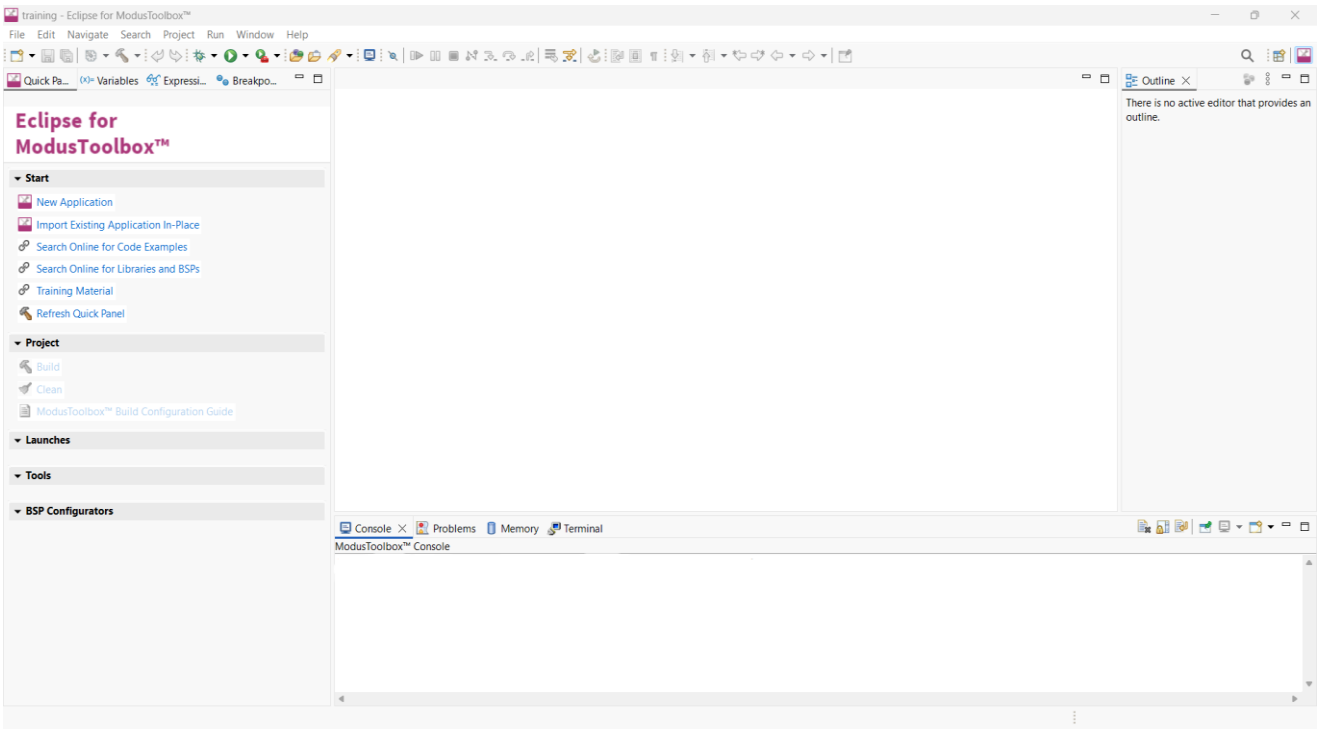


Figure 4 Launch ModusToolbox™ workspace

4. Now select **New Application** from the Quick Panel. Alternatively, go to **File> New> ModusToolbox™** application

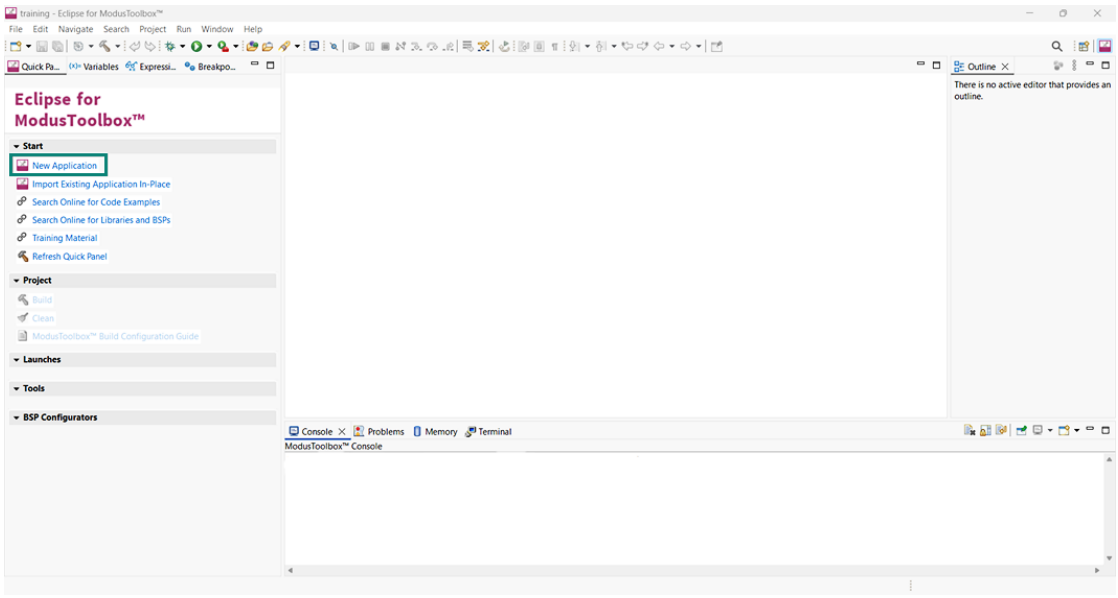


Figure 5 Create new application

5. Clicking New Application opens the **Project Creator Tool** and prompts you to choose a **board support package** (BSP)

*Note: BSPs are aligned with our development/evaluation kits; they provide files for basic device functionality. A BSP typically has a **design.modus** file that configures clocks and other board-specific capabilities. That file is used by the ModusToolbox™ configurators. A BSP also includes the required device support code for the device on the board. You can modify the configuration to suit your application.*

6. Once **Projector Creator** launches, select the **KIT_PSE84_EVAL_EPC2 BSP** under the PSOC™ Edge BSPs and click **Next**

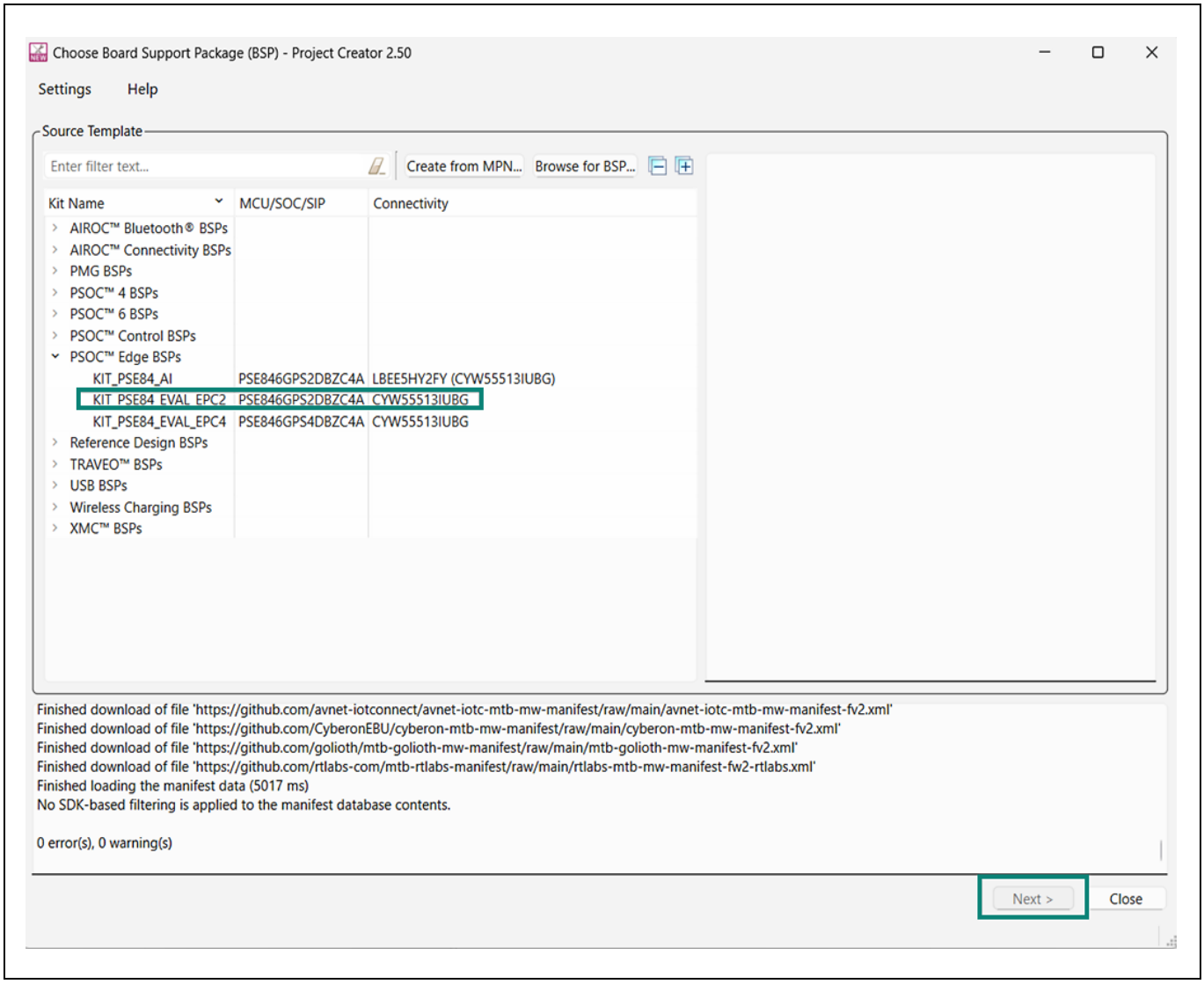


Figure 6 Select the BSP in the project creator

7. Clicking Next takes you to the next menu of the Project Creator, where you can select an application.
   a) Select the **PSOC™ Edge Protect Bootloader** under the **Security** section. You can do this by selecting the **checkbox** near the application
   b) Select the **PSOC™ Edge Basic Secure App** under the **Security** section. You can do this by selecting the **checkbox** near the application
   c) Click **Create**

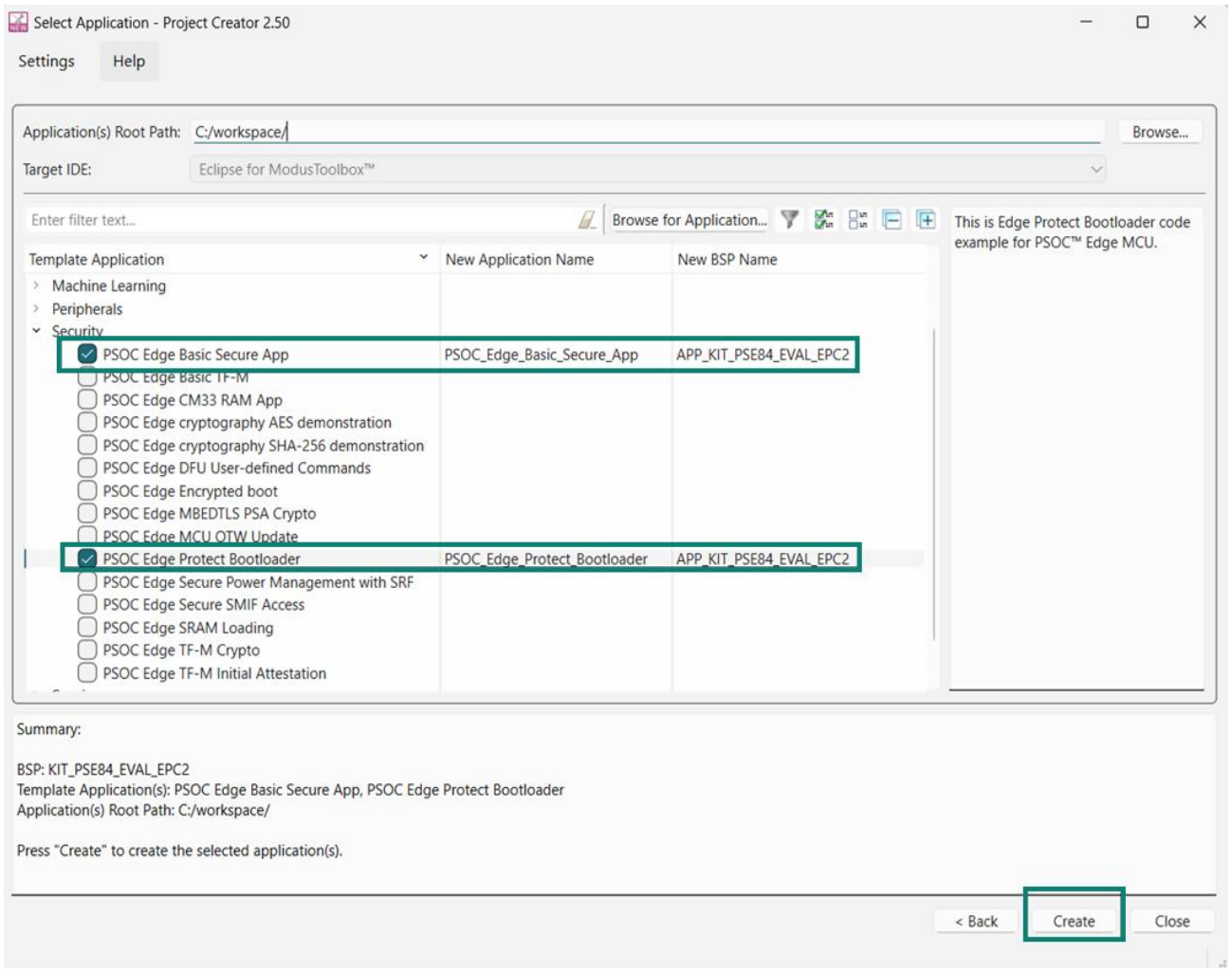


Figure 7 Select application in project creator

Note: Complete demonstration of Edge Protect Bootloader requires the Basic secure application. Creating both code examples together in the above step

8. When you click **Create**, two new ModusToolbox™ applications are created. You can see both the applications in the Project Explorer window as shown in Figure 8

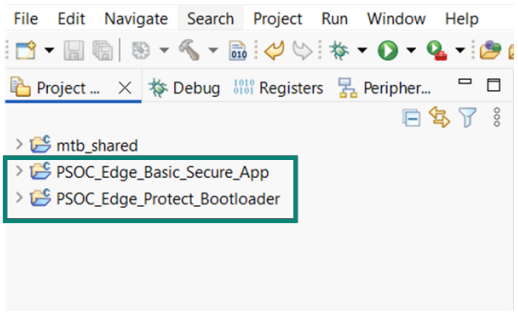


Figure 8 Project explorer window

9. Right-click and copy **proj_bootloader** under the Edge Protect Bootloader

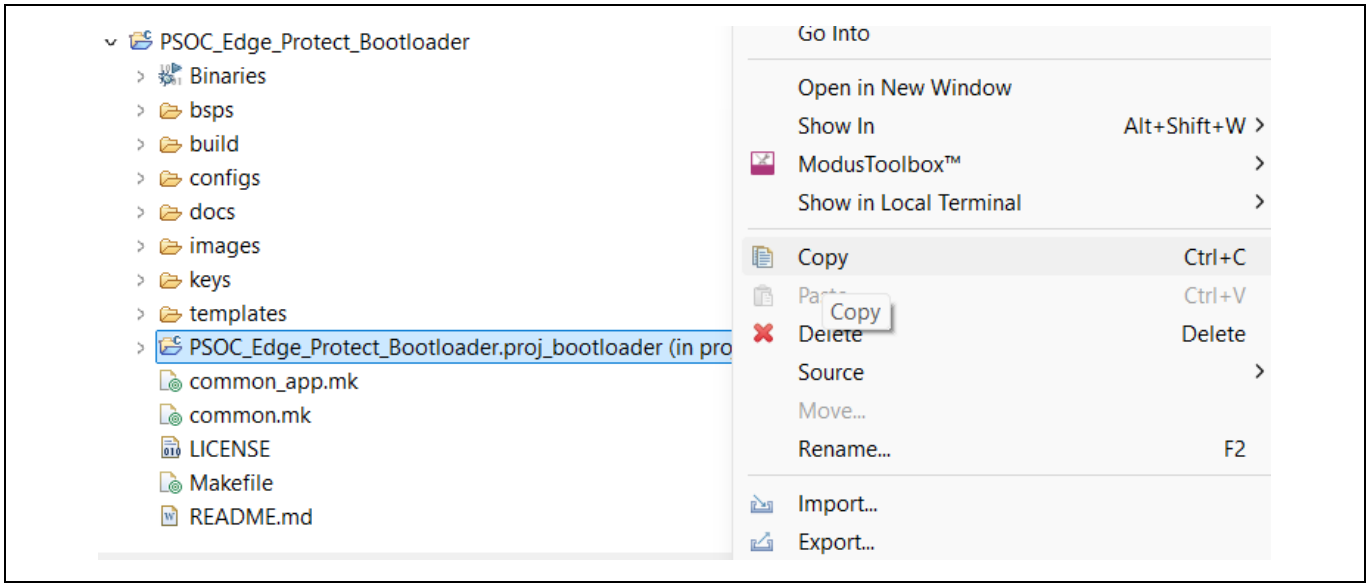


Figure 9 Copy proj_bootloader.

10. Right-click and paste the **proj_bootloader** to the root of Basic Secure App

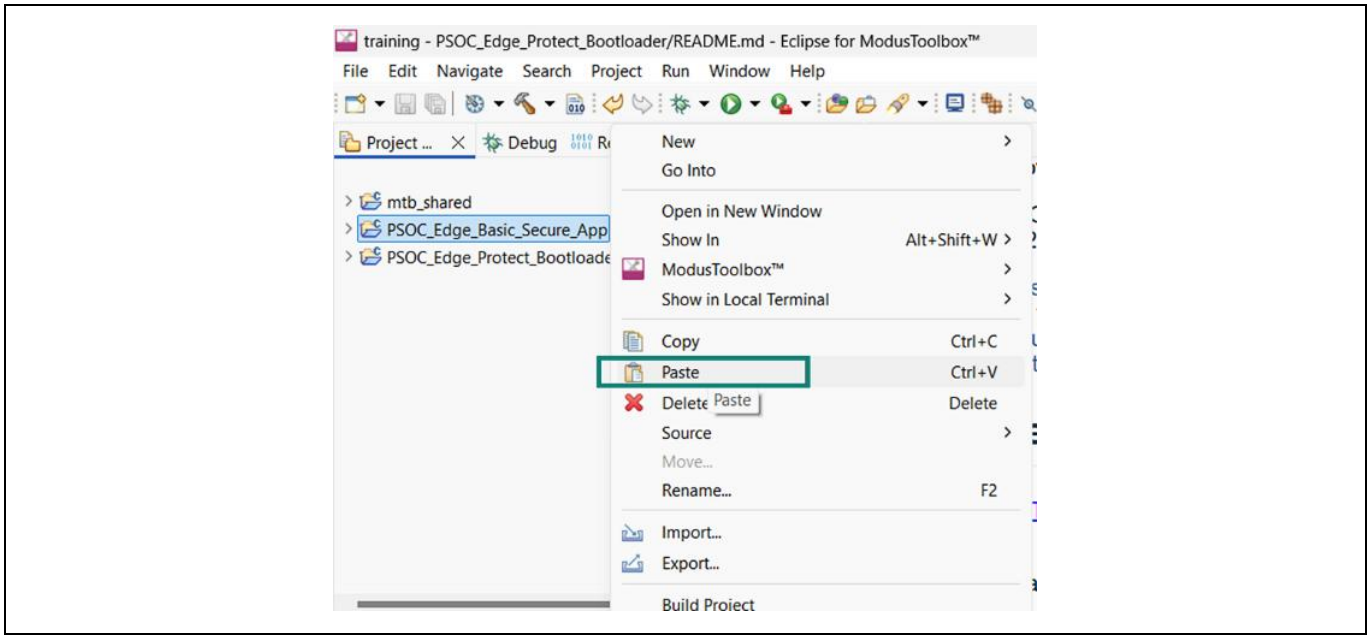


Figure 10 Paste proj_bootloader

11. While pasting, edit the **Project name** and **Location** to point to the Basic Secure App

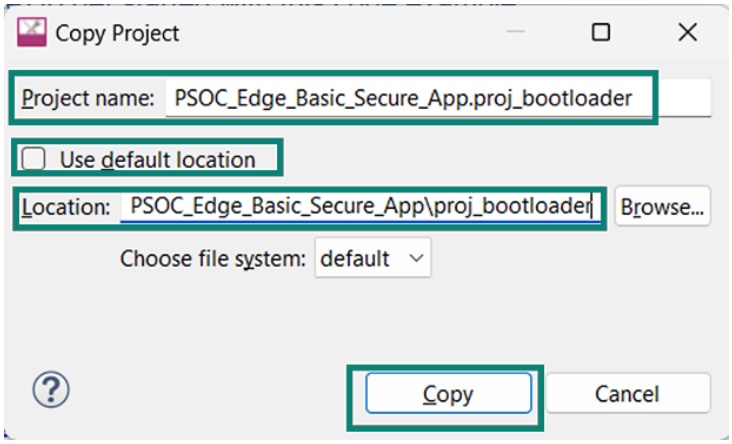


Figure 11 Edit the project name and location

12. Add the **proj_bootloader** to the list of projects in the Makefile of the Basic Secure App

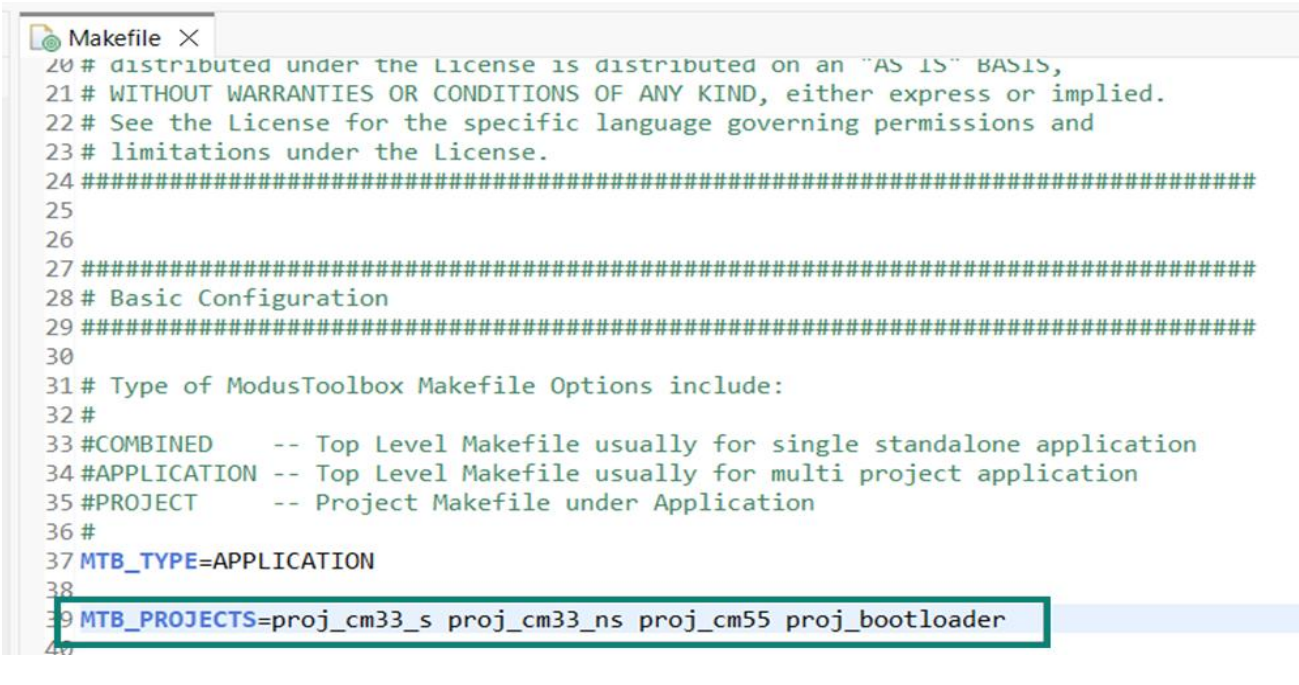


Figure 12 Add proj_bootloader to the ModusToolbox™ projects list

13. Update the common.mk file of the Basic Secure App to point to the bootloader personality generated JSON file

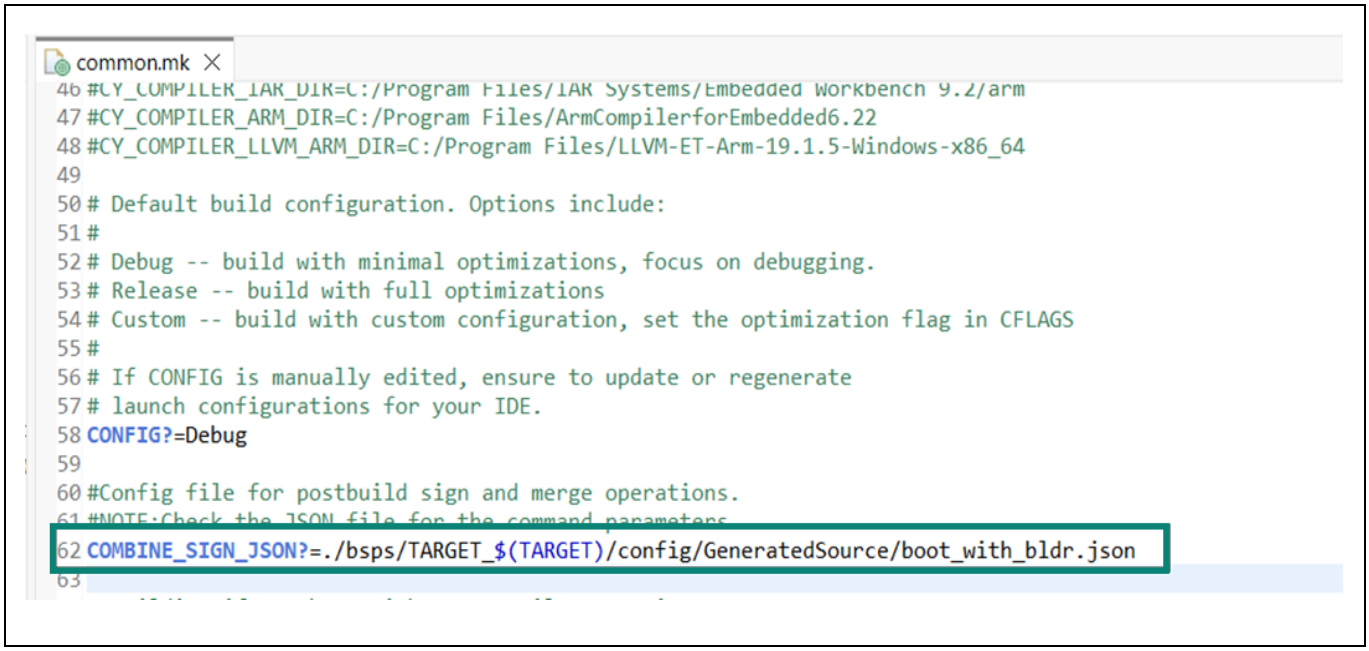


Figure 13 Update JSON file in common.mk

14. Generate launches for the Basic Secure App

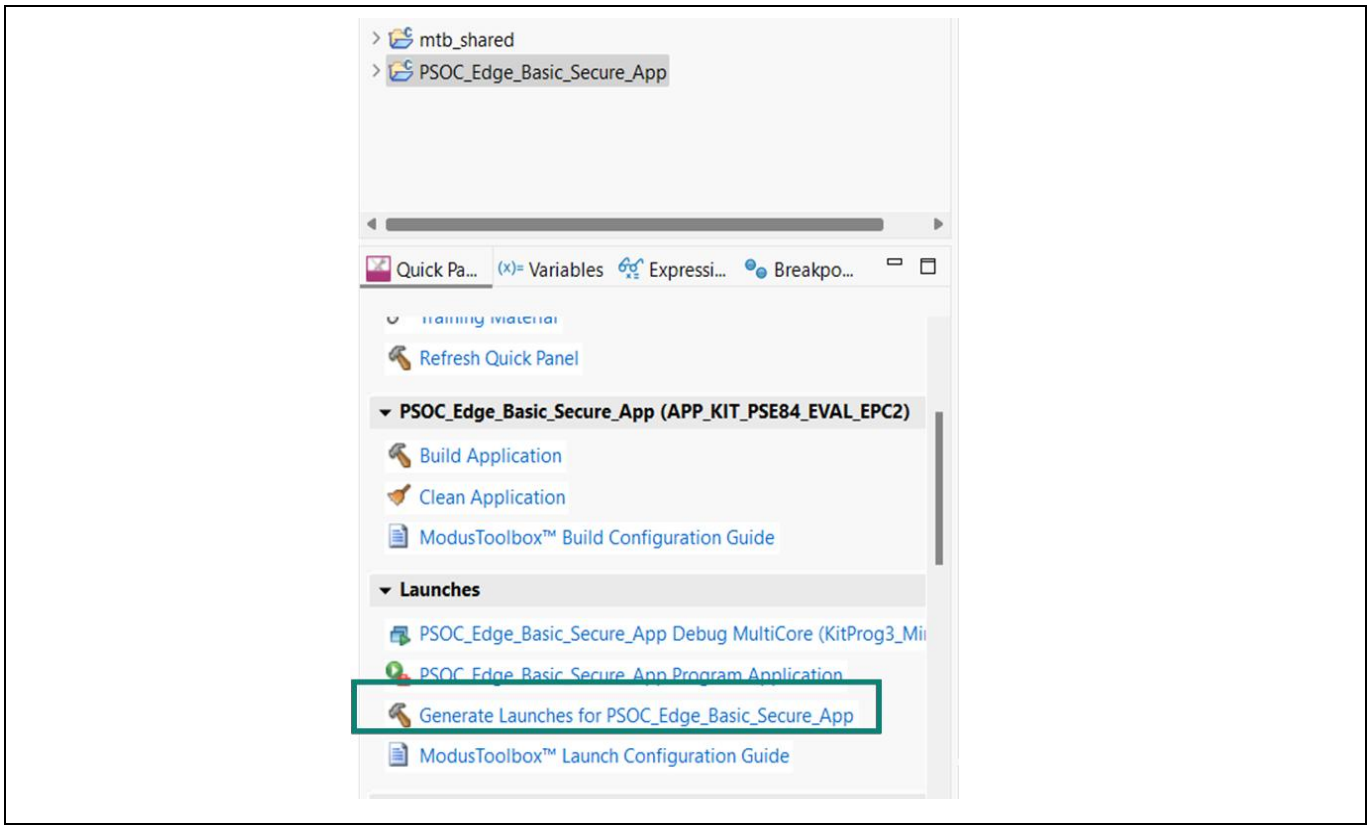


Figure 14 Generate launches for the Basic Secure App

15. Now build and program the Basic Secure App by clicking **Program Application** in the Quick Panel

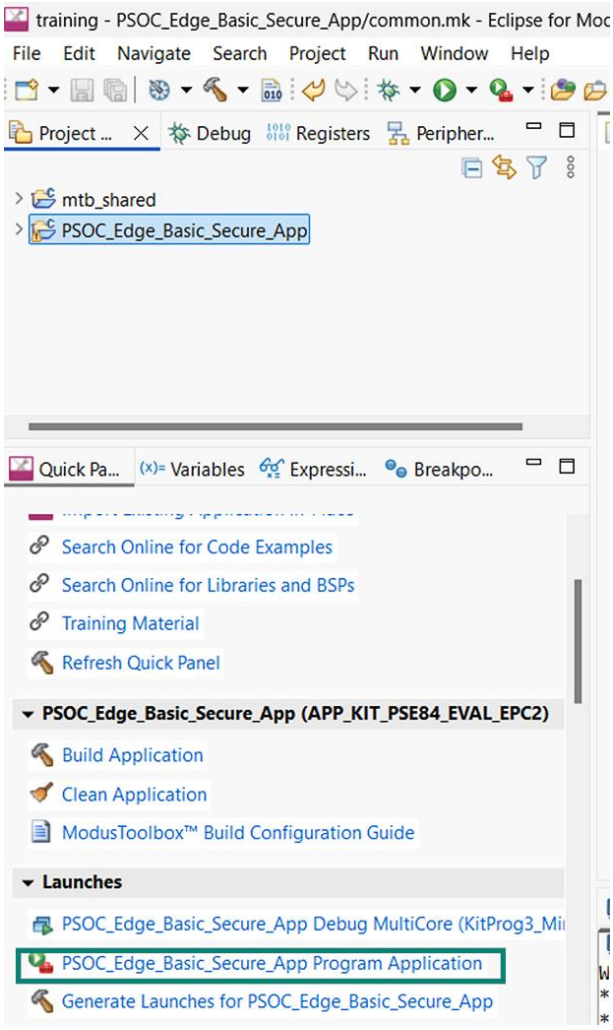


Figure 15 Program application

4.4 Basic Secure App booting to console

1. Set the Boot SW (SW6) / P17.6 position to “LOW / OFF”. This will instruct Extended Boot to boot the first application from RRAM. Edge Protect Bootloader is the first application in this case, which is built and programmed to RRAM in the default case
2. Reset / power cycle the device
3. Observe the Tera Term console logs for Basic Secure App

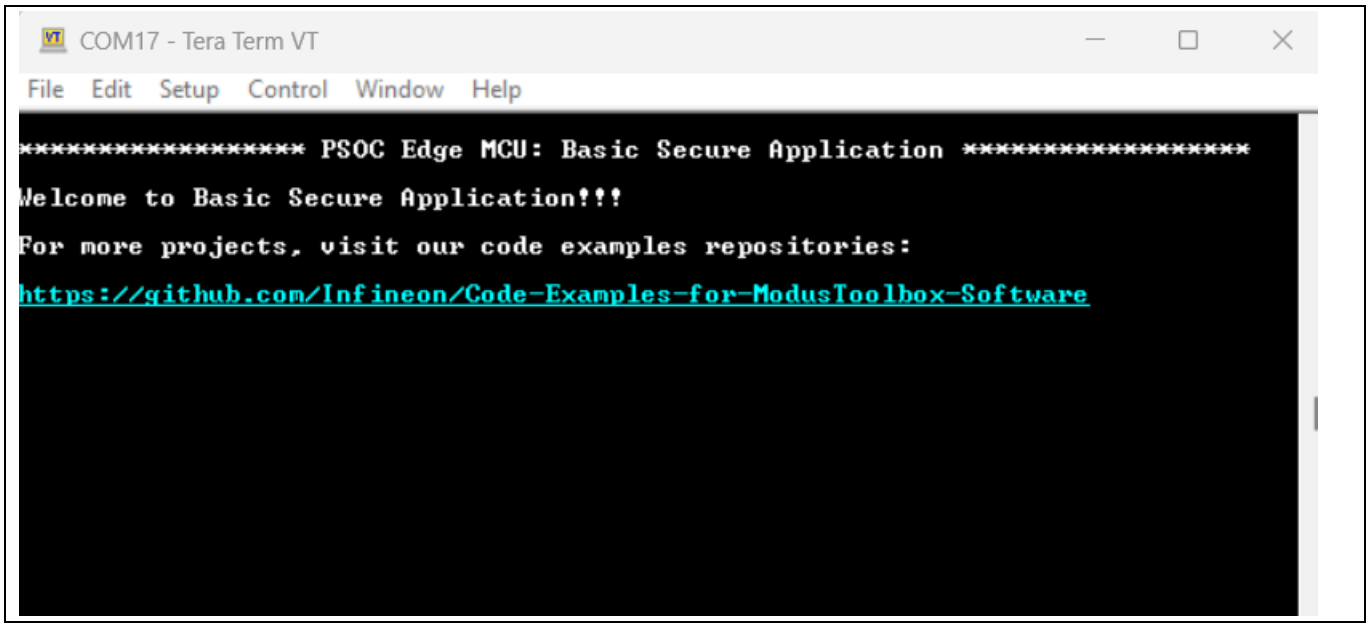


Figure 16 Terminal output for Basic Secure App

4. Observe the Tera Term console logs for the Edge Protect Bootloader before the Basic Secure App logs

Basic Secure app with Edge Protect Bootloader (EPB)



Figure 17 Terminal output for bootloader

4.5 Conclusion

We successfully created and tested the Edge Protect Bootloader for PSOC™ Edge E84.

In the process, we learnt:

- How to add proj_bootloader to the Basic secure application
- How to boot to a validated application using the Edge Protect Bootloader

5 Secure boot with Edge Protect Bootloader

5.1 Objective

Through this lab, you will gain a deeper understanding of secure boot and the chain of trust involved, starting with the Infineon pre-programmed Extended Boot until the user application (Basic Secure App). You will learn how to sign bootloader and application images to extend the Chain of Trust (CoT).

5.2 Description

Enabling the complete chain of trust requires Extended Boot to verify Edge Protect Bootloader, and Edge Protect Bootloader should verify the user application (all projects in the user application). To do that, you will have to do the following:

1. Obtain the device ownership
2. Enable secure boot in the OEM policy
3. Configure the bootloader solution personality to enable image validation
4. Build and program the user application

5.3 Project Creation

The project creation process is the same as discussed in the section 4.3.

5.4 Enabling Secure Boot

5.4.1 Obtain the device ownership

The process for transferring device ownership is detailed in the "Training Manual PSOC™ Edge E84 Getting Started with Security". Adhere to the precise instructions outlined in the manual to generate the key pair and provision the device using edgeprotecttools to obtain device ownership. If you have previously obtained device ownership through this process, you can skip this step.

It is important to note that the same key pair (OEM_RoT_Key) used for provisioning must be used for signing and verification.

5.4.2 Enable Secure Boot in policy

Secure boot is disabled by default. You can enable Secure Boot by modifying the default policy and provisioning it to the device. Secure boots must be enabled first in the policy, and the new policy must be provisioned

1. Edit <application-directory>/policy/policy_oem_provisioning.json file using any text editor to set the value field to true for the “secure_boot” parameter

```
"extended_boot_policy": {
  "secure_boot": {
    "description": "Disable/Enable secure boot option",
    "value": true
  },
```

Figure 18 Extended boot policy

2. Connect the board to the host machine (PC) using a USB cable
3. In modus-shell, navigate to <application-directory> and execute the following command to provision the device with the updated policy

```
edgeprotecttools -t pse8xs2 provModision-device -p policy/policy_oem_provisioning.json --key keys/oem_private_key_0.pem
```

For further details**,** see the **"**Enable Secure Boot in Extended Boot" section from the "PSOC™ Edge E84 Training manual for Security Introduction" lab.

5.4.3 Configure the Edge Protect Bootloader for image validation

In the Enabling Secure Boot section, we enabled the Secure Boot capability in Extended Boot. The Extended boot will now validate the signature of the next image (Edge Protect Bootloader in this case) before booting it.

You will have to sign the Edge Protect Bootloader image. Additionally, to configure the Edge Protect Bootloader to validate the subsequent images.

Follow the steps below to enable EPB image validation and to sign the EPB image:

1. Ensure the oem_private_key_0.*pem* (pair of OEM_RoT_Key Public provisioned to the device) is available in the Basic_Secure_App/keys directory
2. Open the device configurator for Basic Secure App

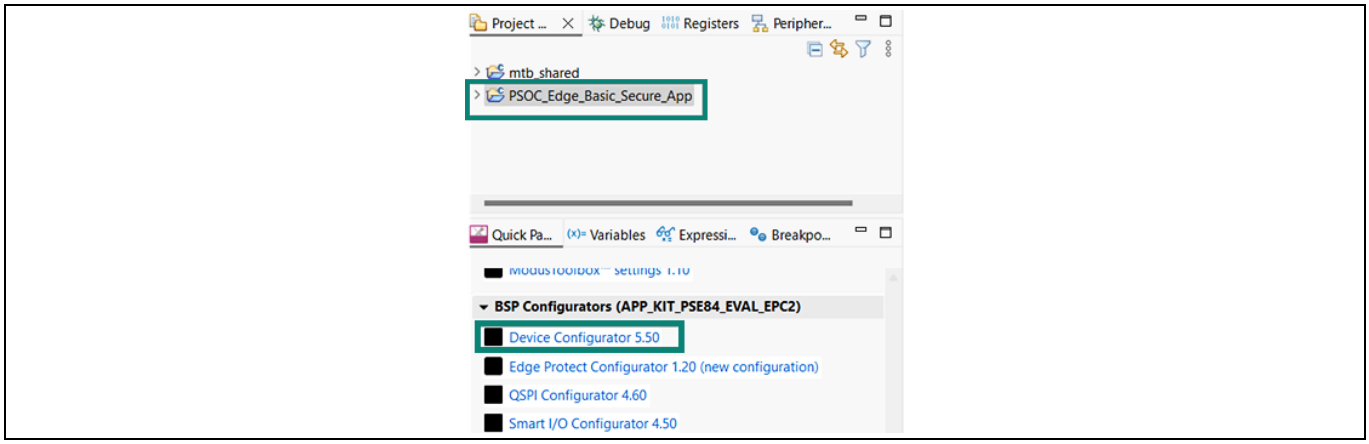


Figure 19 Open Device Configurator

3. Go to the **Solutions** tab of the Device configurator, under the **Parameters** view of the **Edge Protect Bootloader solution,** and enable the following:
   a) Validate boot
   b) Validate upgrade
   c) Enable bootloader signing (this option enables signing of the bootloader image)

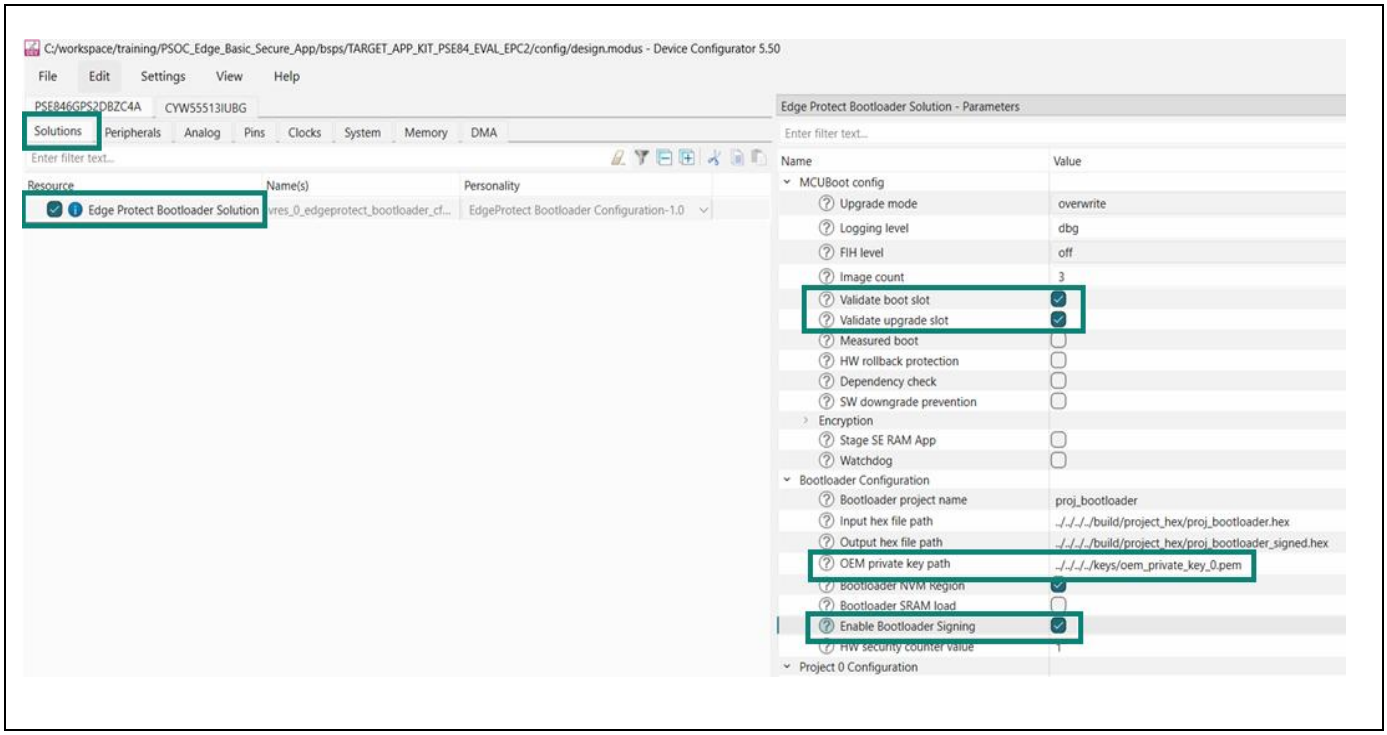


Figure 20 Enable image validation.

4. Save your changes to the **design.modus** to generate bootloader configuration files
5. Now build and program the Basic Secure App by clicking the **Program Application** button in the Quick Panel

5.5 Conclusion

- We successfully enabled Secure Boot of Edge Protect Bootloader and the user application, thereby establishing a chain of trust(CoT) from Infineon programmed software to the user application
- We learnt how to enable Secure Boot capability in Extended boot and Edge Protect Bootloader and how to sign each of these images

Revision history

Document revision	Date	Description of changes
**	2025-11-13	Initial revision
*A	2026-05-08	Fixed formatting. Updated figures to align with brand guidelines.

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